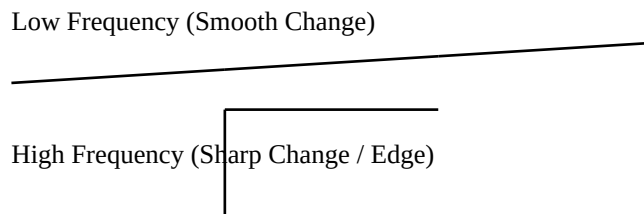


Fourier Transform and Hough Transform (BS Level Notes)

1. Fourier Transform

Fourier Transform converts an image from Spatial Domain (pixel values) to Frequency Domain (rate of intensity change). Low Frequency: Smooth areas (slow intensity change) High Frequency: Edges and sharp changes Applications: - Image Filtering - Noise Removal - JPEG Compression - Pattern Analysis

Low vs High Frequency Concept

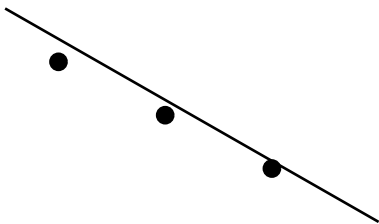


2. Hough Transform

Hough Transform is used to detect geometric shapes such as lines and circles. Steps for Line Detection: 1. Detect edges (e.g., Canny Edge Detector) 2. Each edge point votes for possible lines 3. Intersection of votes gives detected line Line Equation (Polar Form): $\rho = x \cos(\theta) + y \sin(\theta)$ Applications: - Lane Detection - Coin Detection - Iris Detection - Industrial Shape Detection

Line Detection Concept

Edge Points in Image Space



Voting in Hough Space

